

1 Theory

A counter is a sequential logic circuit that counts pulses or events, often used in digital systems for timing, frequency measurement, or data counting. It consists of a series of flip-flops connected in such a way that their states change in response to input clock pulses. Counters can be classified into various types based on their functionality, such as up counters, down counters, and up-down counters. These counters are widely used in digital electronics for tasks like tracking the number of occurrences or sequencing operations. There are mainly three common types of 4-bit counters: up counter, down counter, and up down counter.

4-Bit Up Counter

A 4-bit up counter is a sequential circuit that counts upward in binary from 0000 to 1111 and then wraps around back to 0000. Each clock pulse increments the binary value stored in the counter. The counter comprises four flip-flops, typically configured as T or JK flip-flops, where each flip-flop represents one bit. The output of one flip-flop is used to toggle the next flip-flop in sequence. As a result, the least significant bit (LSB) toggles on every clock pulse, and higher bits toggle at frequencies proportional to their position in the binary sequence. The 4-bit up counter is commonly used in timing circuits, digital clocks, and frequency dividers.

4-Bit Down Counter

A 4-bit down counter works similarly to an up counter, but it counts in reverse order, decrementing the binary value with each clock pulse. Starting from 1111, the counter decreases sequentially to 0000 and then wraps back to 1111. Like the up counter, a down counter also uses four flip-flops, with the difference being in the design of the toggling logic, ensuring that the counter counts downward. The outputs of the flip-flops are arranged to decrement the value. Applications of 4-bit down counters include countdown timers, digital clocks, and reverse sequencing operations.

4-Bit Up Down Counter

A 4-bit up-down counter is a versatile counter that can operate in both up-counting and down-counting modes, depending on a control signal. The mode control input determines the direction of counting, allowing the counter to increment or decrement based on the clock pulses. Internally, the counter uses additional logic to toggle the states of the flip-flops accordingly. The up-down counter finds applications in bidirectional counting scenarios, such as elevators, reversible stepper motors, or systems requiring dynamic counting in both directions. This dual functionality makes the up-down counter highly adaptable in digital system design.

2 Implementation and Observation

2.1 4-Bit Up Counter

VHDL code

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_signed.all;

entity counter is
port (clk:in std_logic;
reset: in std_logic;
count: out std_logic_vector(3 downto 0);
load: in std_logic);
end counter;

architecture behavioural of counter is
begin
process(clk, reset, load)
variable temp :std_logic_vector(3 downto 0):="0000";
begin
if (reset='1') then
temp:="0000";
elsif (rising_edge(clk)) then
if(load='1') then
temp:=temp+1;
end if;
end if;
count<= temp;
end process;

end behavioural;
```

Test Bench

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity counter_tb is
-- Port ();
end counter_tb;

architecture Behavioural of counter_tb is component counter is
port (clk:in std_logic;
```

```

reset: in std_logic;
count: out std_logic_vector(3 downto 0);
load: in std_logic;
end component;

signal clk: std_logic := '0';
signal reset: std_logic := '0';
signal load: std_logic := '0';
signal count: std_logic_vector(3 downto 0);

begin
uut: counter port map (
clk => clk,
reset => reset,
load => load,
count => count
);

clk_process: process
begin
clk <=not clk;
wait for 10 ns;
end process;

process
begin
load <='1' ; reset <='0' ;
wait for 100 ns;

load <='0' ; reset <= '0' ;
wait for 100 ns;

load <= '1';
reset <= '0';
wait for 100 ns;

load <='1';
reset <= '0';
wait for 100 ns;

wait;
end process;

end Behavioural;

```

RTL Schematic Diagram

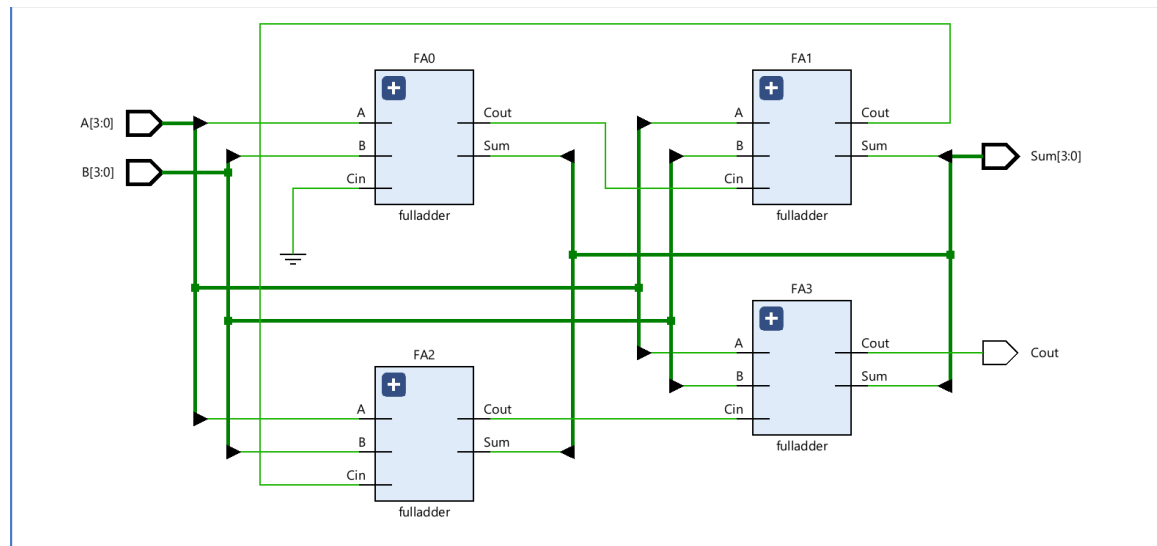


Figure 1: RTL Schematic Diagram of 4-bit up counter

RTL design is used to describe behavior of the 4-bit up counter as shown in figure 1.

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the 4-bit up counter circuit as shown in figure 2.

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given theoretical and practical explanation of 4-bit up counter as shown in figure 3.

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 4.

Power Estimation

Following figure shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a 4-bit up counter circuit as shown in figure 5.

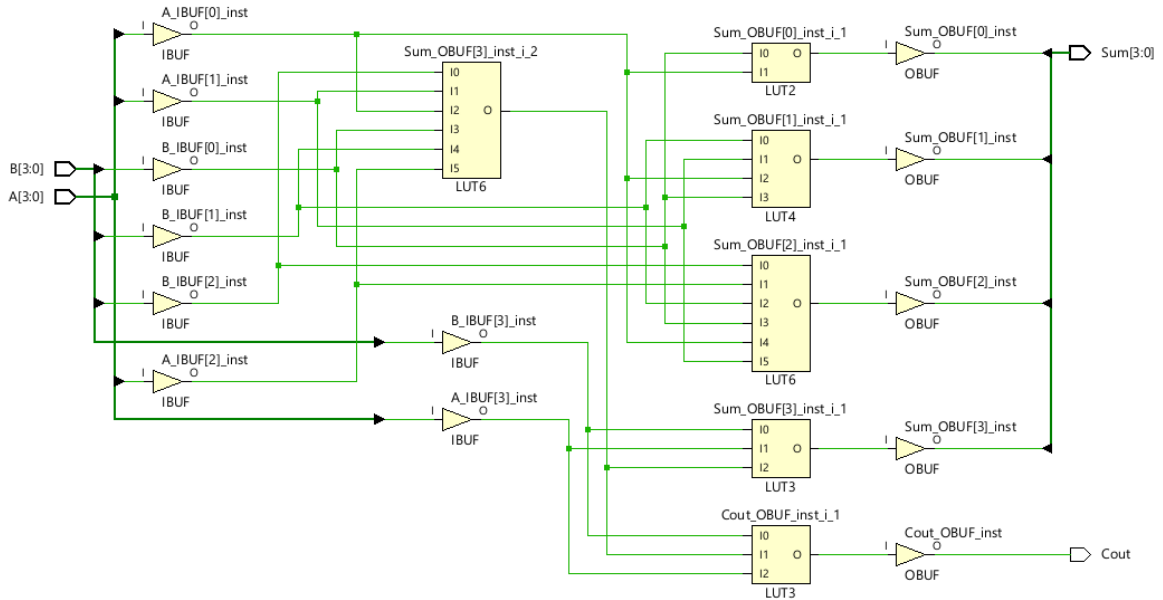


Figure 2: Synthesis Schematic Diagram of 4-bit up counter

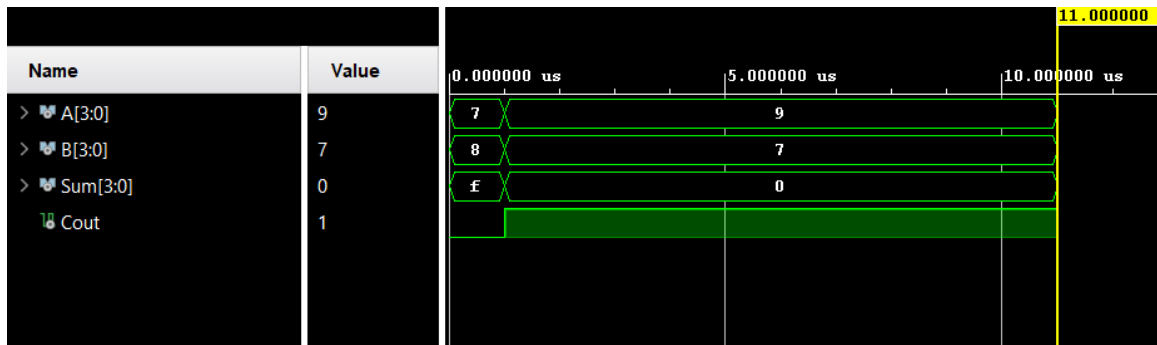


Figure 3: Timing Diagram of 4-bit up counter

2.2 4-Bit Down Counter

VHDL code

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_signed.all;

entity counter is
port (clk:in std_logic;
reset: in std_logic;
count: out std_logic_vector(3 downto 0);
load: in std_logic);
end counter;
```

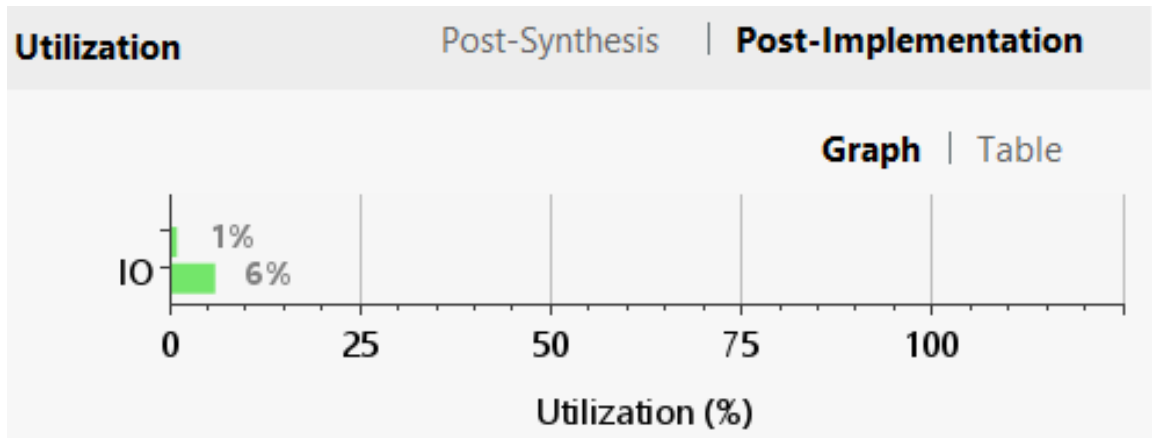


Figure 4: Device Utilization of 4-bit up counter

```

architecture behavioural of counter is
begin
process(clk, reset, load)
variable temp :std_logic_vector(3 downto 0):="0000";
begin
if (reset='1') then
temp:="0000";
elsif (rising_edge(clk)) then
if(load='1') then
temp:=temp-1;
end if;
end if;
count<= temp;
end process;

end behavioural;

```

Test Bench

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity counter_tb is
-- Port ();
end counter_tb;

architecture Behavioural of counter_tb is component counter is
port (clk:in std_logic;
reset: in std_logic;

```

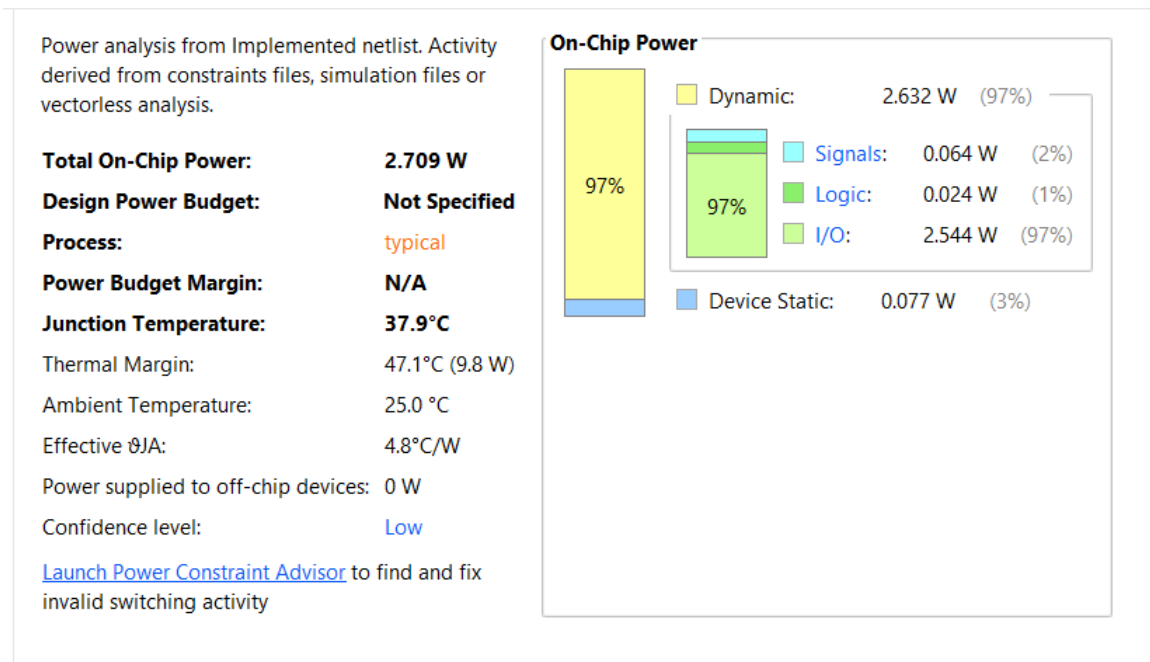


Figure 5: Power Estimation of 4-bit up counter

```

count: out std_logic_vector(3 downto 0);
load: in std_logic);
end component;

signal clk: std_logic := '0';
signal reset: std_logic := '0';
signal load: std_logic := '0';
signal count: std_logic_vector(3 downto 0);

begin
  uut: counter port map (
    clk => clk,
    reset => reset,
    load => load,
    count => count
  );

  clk_process: process
  begin
    clk <= not clk;
    wait for 10 ns;
  end process;

  process

```

```

begin
load <='1' ; reset <='0' ;
wait for 100 ns;

load <='0' ; reset <= '0' ;
wait for 100 ns;

load <= '1';
reset <= '0';
wait for 100 ns;

load <='1';
reset <= '0';
wait for 100 ns;

wait;
end process;

end Behavioural;

```

RTL Schematic Diagram

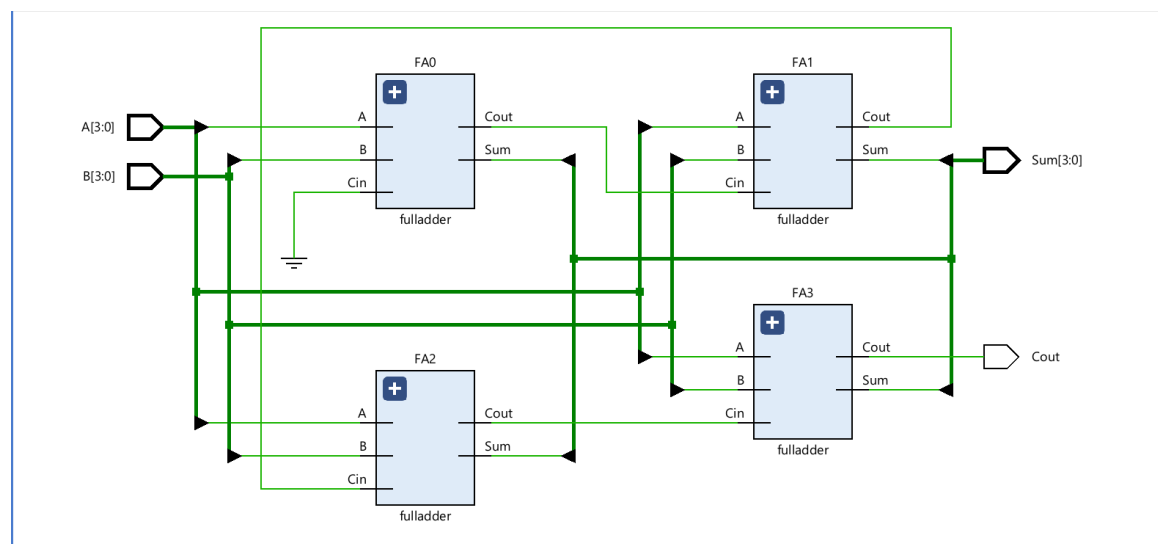


Figure 6: RTL Schematic Diagram of 4-bit down counter

RTL design is used to describe behavior of the 4-bit down counter as shown in figure 6.

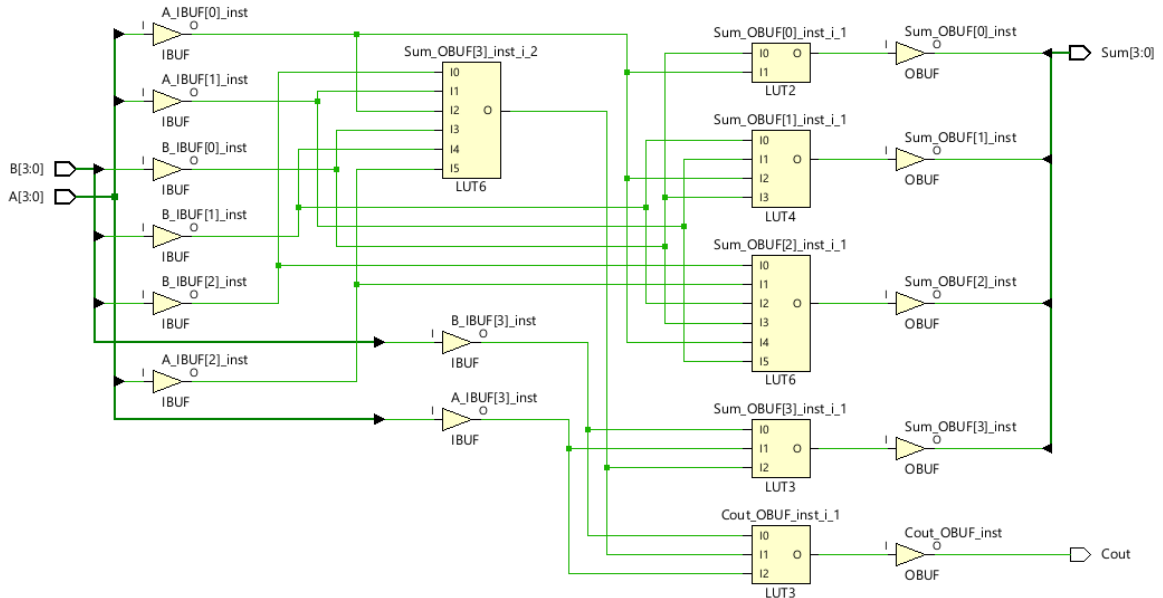


Figure 7: Synthesis Schematic Diagram of 4-bit down counter

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the 4-bit down counter circuit as shown in figure 7.

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given theoretical and practical explanation of 4-bit down counter as shown in figure 8.

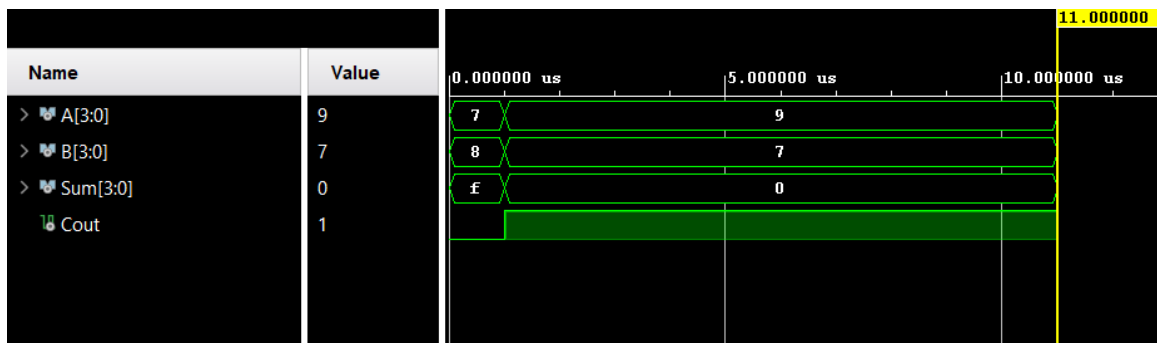


Figure 8: Timing Diagram of 4-bit down counter

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on

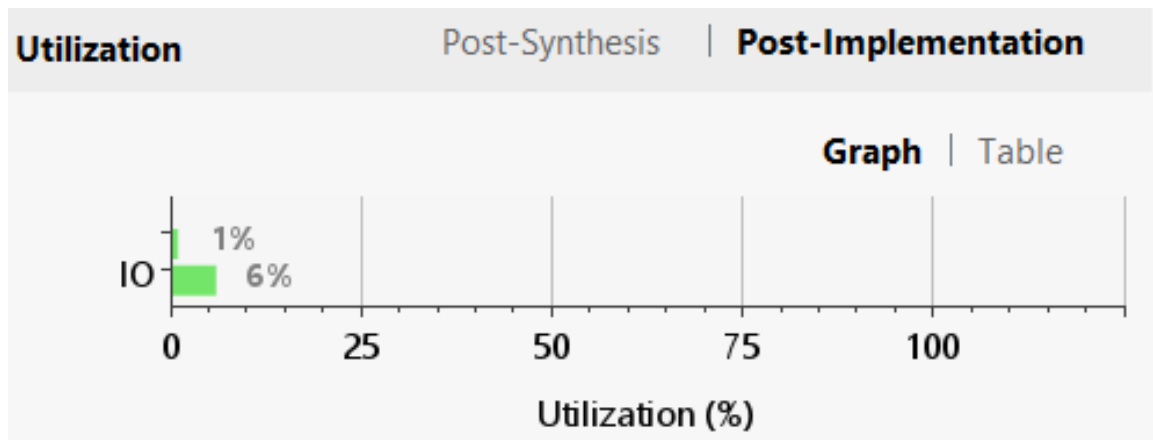


Figure 9: Device Utilization of 4-bit down counter

the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 9.

Power Estimation

Following figure shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a 4-bit down counter circuit as shown in figure 10.

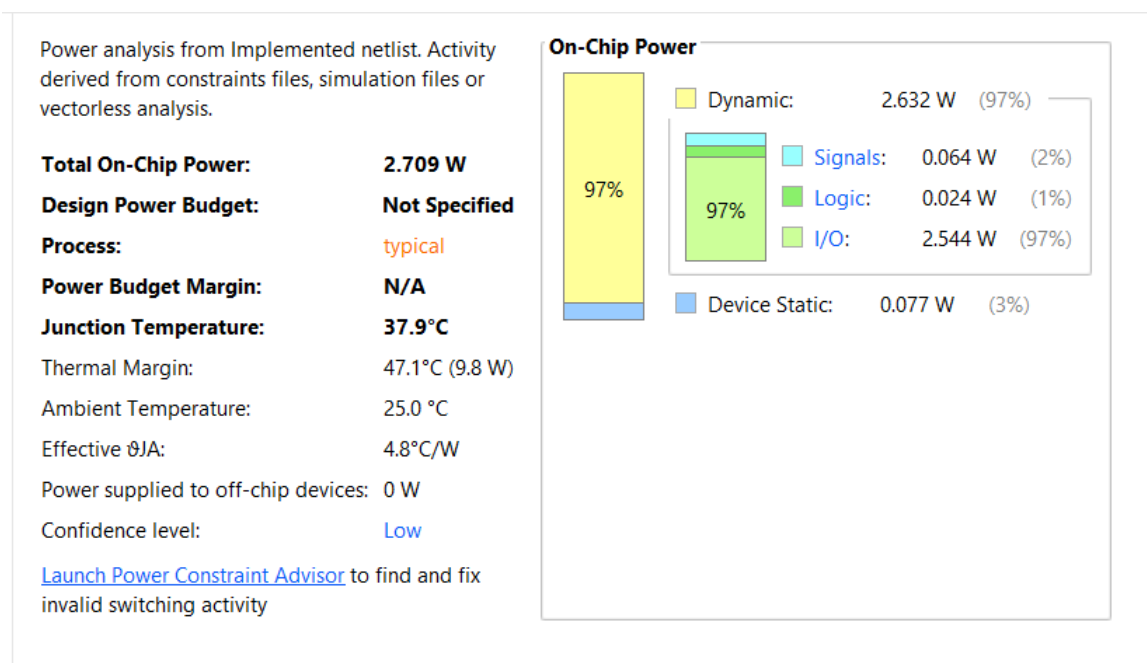


Figure 10: Power Estimation of 4-bit down counter

2.3 4-Bit Up Down Counter

VHDL code

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_signed.all;

entity counter is
port (clk: in_std_logic;
reset: in std_logic;
count: out std_logic_vector(3 downto 0);
m: in std_logic;
load: in std_logic);
end counter;

architecture behavioural of counter is
begin
process (clk, reset, load)
variable temp : std logic vector( vector (3 downto 0):="0000";

begin
if (reset='1') then
temp:="0000";
elsif (rising_edge(clk)) then
if (load='1') then
if(m = '0') then
temp:=temp+1;
else
temp:=temp-1;
end if;
end if;
end if;

count<= temp;
end process;

end behavioural;
```

Test Bench

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity counter_tb is
```

```

-- Port ();
end counter_tb;

architecture Behavioural of counter_tb is component counter is
port (clk:in std_logic;
reset: in std_logic;
count: out std_logic_vector(3 downto 0);
load: in std_logic;
m: in std_logic);
end component;

signal clk: std_logic := '0';
signal reset: std_logic := '0';
signal load: std_logic := '0';
signal count: std_logic_vector(3 downto 0);
signal m: std_logic := '0';

begin
ut: counter port map (
clk => clk,
reset => reset,
load => load,
count => count
);

clk_process: process
begin
clk <=not clk;
wait for 10 ns;
end process;

process
begin
load <='1' ;
reset <='0' ;
m = 0;
wait for 100 ns;

load <='0' ; reset <= '0' ;
wait for 100 ns;

load <= '1';
reset <= '0';
m <= 1;
wait for 100 ns;

```

```

load <='1';
reset <= '0';
wait for 100 ns;

wait;
end process;

end Behavioural;

```

RTL Schematic Diagram

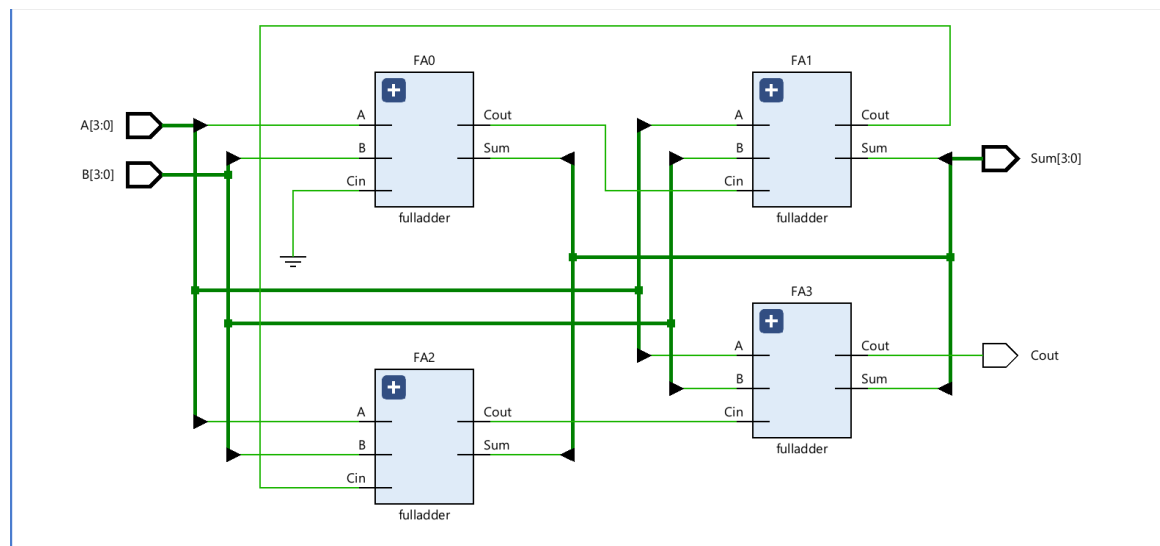


Figure 11: RTL Schematic Diagram of 4-bit up down counter

RTL design is used to describe behavior of the 4-bit up down counter as shown in figure 11.

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the 4-bit up down counter circuit as shown in figure 12.

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given theoretical and practical explanation of 4-bit up down counter as shown in figure 13.

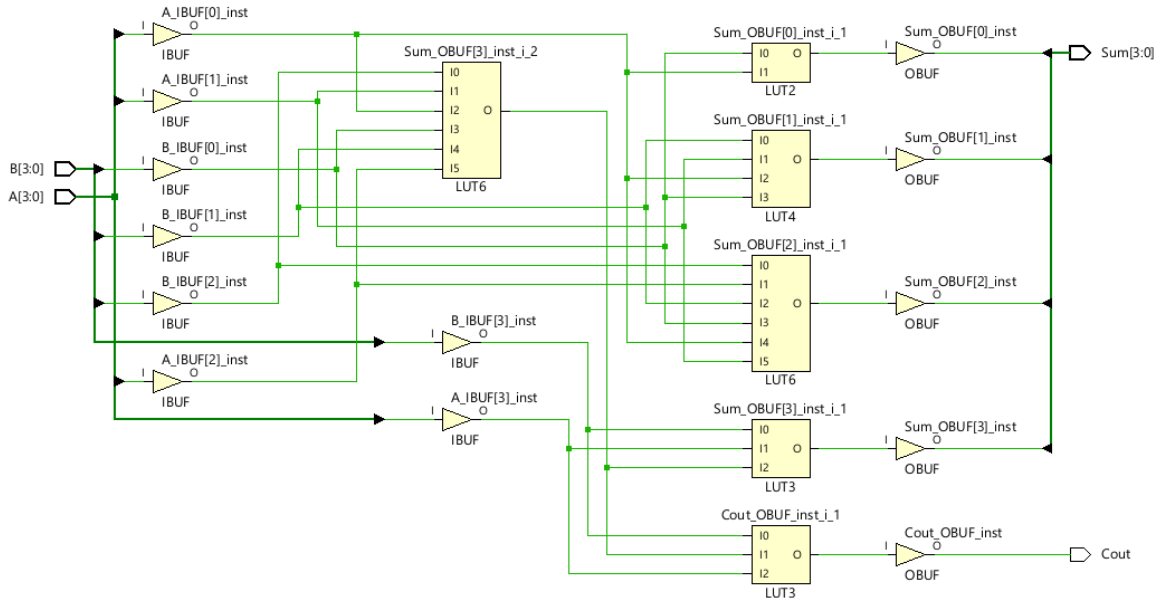


Figure 12: Synthesis Schematic Diagram of 4-bit up down counter

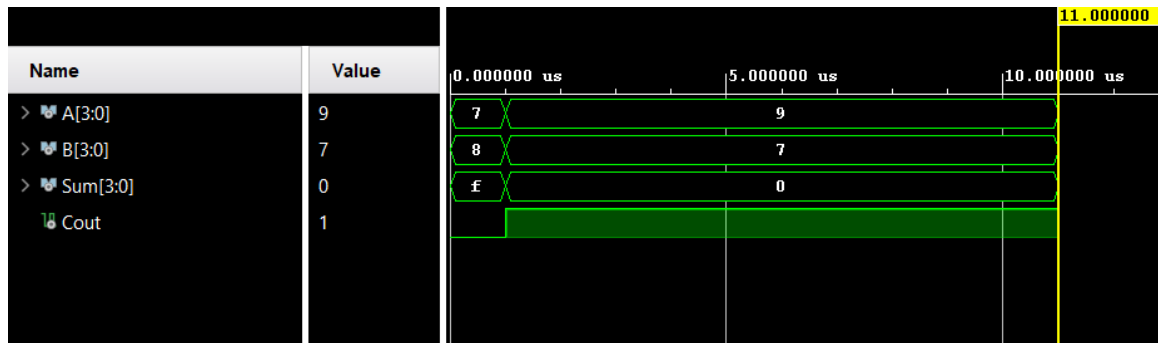


Figure 13: Timing Diagram of 4-bit up down counter

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 14.

Power Estimation

Following figure shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a 4-bit down counter circuit as shown in figure 15.

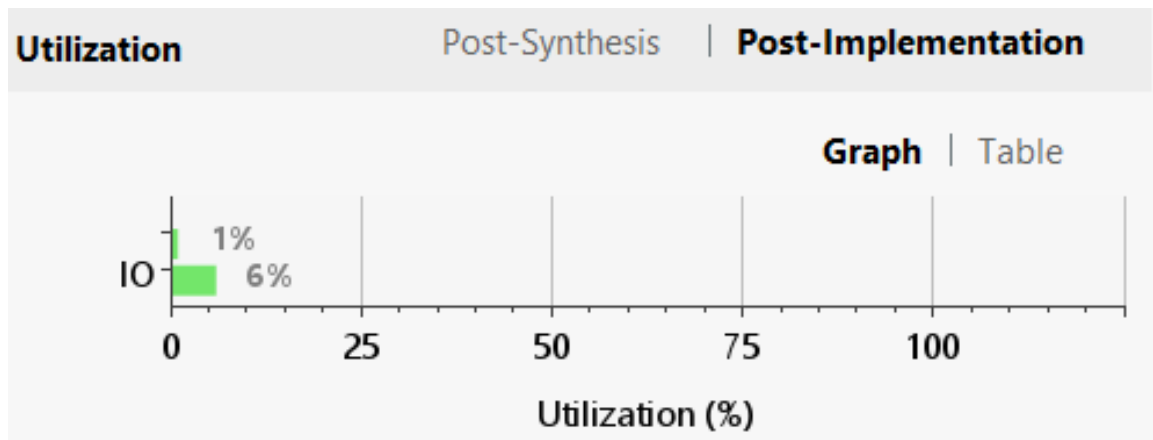


Figure 14: Device Utilization of 4-bit up down counter

3 Conclusion

In this lab, we implemented the VHDL code for four bit - up counter, down counter and up-down counter on the Atrix-7 FPGA (part: xc7a50tcsg324-1, package: csg324) using Vivado software. Each design was synthesized, simulated, and implemented, which confirmed their correct functionality.

The process began with simulation to verify the timing diagrams, followed by linting to check for syntax errors. Afterward, the designs were synthesized into a hardware logic circuit, and during the implementation phase, power and timing characteristics were analyzed, displayed and studied.

The implementation and simulation of 4-bit up, down, and up-down counters in Vivado software provided valuable insights into the design and behavior of digital counters. Each counter type demonstrated its unique functionality, with the up counter incrementing sequentially, the down counter decrementing, and the up-down counter offering bidirectional counting based on control signals.

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: **2.709 W**
Design Power Budget: **Not Specified**
Process: typical
Power Budget Margin: N/A
Junction Temperature: **37.9°C**
Thermal Margin: 47.1°C (9.8 W)
Ambient Temperature: 25.0 °C
Effective θ_{JA} : 4.8°C/W
Power supplied to off-chip devices: 0 W
Confidence level: Low
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power

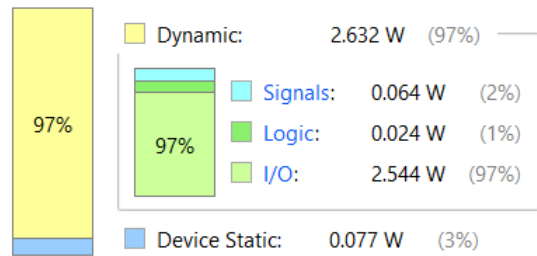


Figure 15: Power Estimation of 4-bit down counter